Exp no: 4 SETTING UP A LOCAL Area NETWORK (LAN)

1. Aim:

To design, set up, and configure a LAN connecting four computers through a network switch, assign static IP address to all devices, verify connectivity using ping command and demonstrate resource/data sharing across the world.

2. Equipment Required:

S.No	Equipment	Quantity	Specification
1.	Personal computers (PC)	4	Ethernet enabled
2.	Network switch	1	8/16/24 port, unmanaged/managed
3.	Ethernet cables	4	straight, RJ-45 connectors

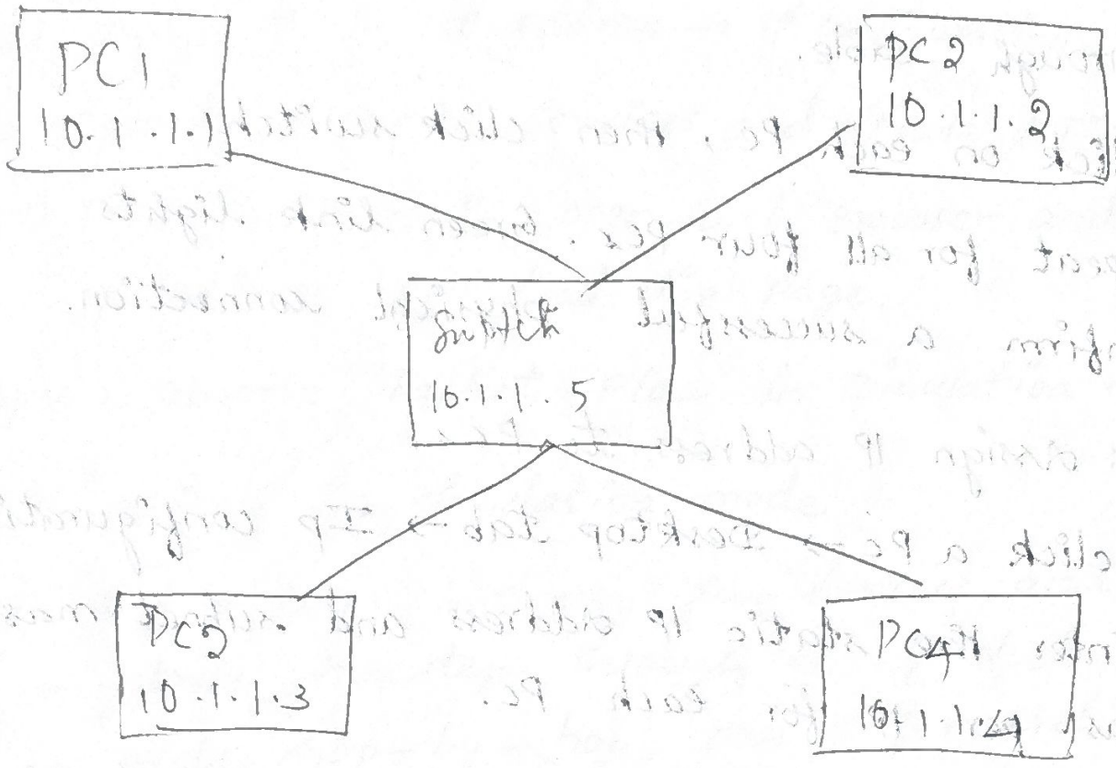
3. Theory:

A local Area Network is a Network that connects computers and devices within a limited geographical area such as home, office or laboratory allowing them to share resources such as files, printers and internet connections.

A star Topology is used in this experiment, where all end devices are connected to a central network switch. Every device on a LAN must be assigned an IP address within same network range to communicate with each other.

4. Network Diagram:

The LAN was implemented using star Topology as shown below, with 4 PC's connected individually to a central switch.



5. IP Address configuration Table:

Device	IP Address	Subnet Mask	Connected To
PC0	10.1.1.1	255.0.0.0	Switch Port Fa0/1
PC1	10.1.2	255.0.0.0	Switch Port Fa0/2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa0/4
Switch	10.1.1.5	255.0.0.0	VLAN 1 interface

6. Implementation in cisco Packet Tracer:

Since physical hardware was limited, the LAN was also implemented and verified using cisco Packet Tracer.

Step 1 : Place the devices

1) Open cisco Packet Tracer
2) From the device category panel, select End devices.
3) Select switches and drag 1 switch.

step 2 : Cable the Devices

4) click connections → Select copper straight-Through cable.

5) click on each Pc, then click switch

6) Repeat for all four Pcs. Green link lights confirm a successful physical connection.

step 3 : Assign IP address to Pcs

7. click a Pc → Desktop tab → Ip configuration

8. Enter the static IP Address and subnet mask as per IP for each Pc.

step 4 : Configure the switch management IP (optional)

9. click the switch → CLI tab

10. Enter the following commands :

configure terminal
interface vlan1
Ip address 10.1.1.5 255.0.0.0
no shutdown
exit

step 5 : Verify connectivity Using Ping

11. click a Pc → desktop → Command prompt

12. Use the ping command to test connectivity

13. Successful "Reply from" messages confirm connectivity.

step6: Demonstrate Data sharing

14) Add a server device

15) Assign it in IP address → IP configuration tab

16) Open the server's services and enable HTTP service.

17) From another PC, open web Browser and enter IP address to load the page.

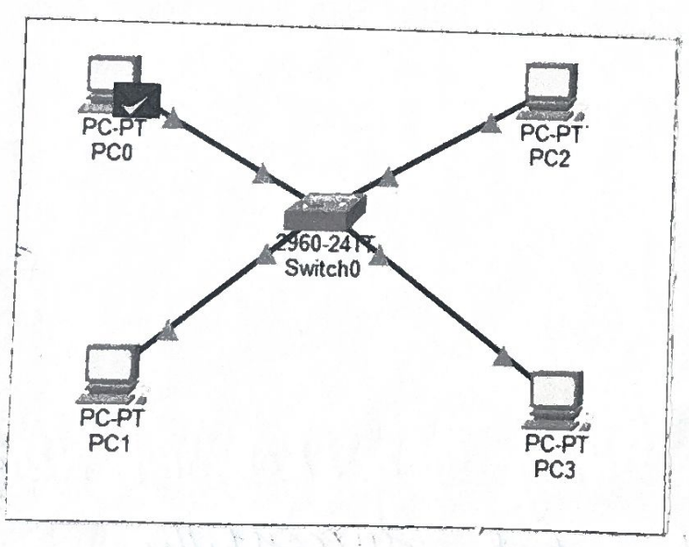
step7: Observe Packet Flow in simulation mode

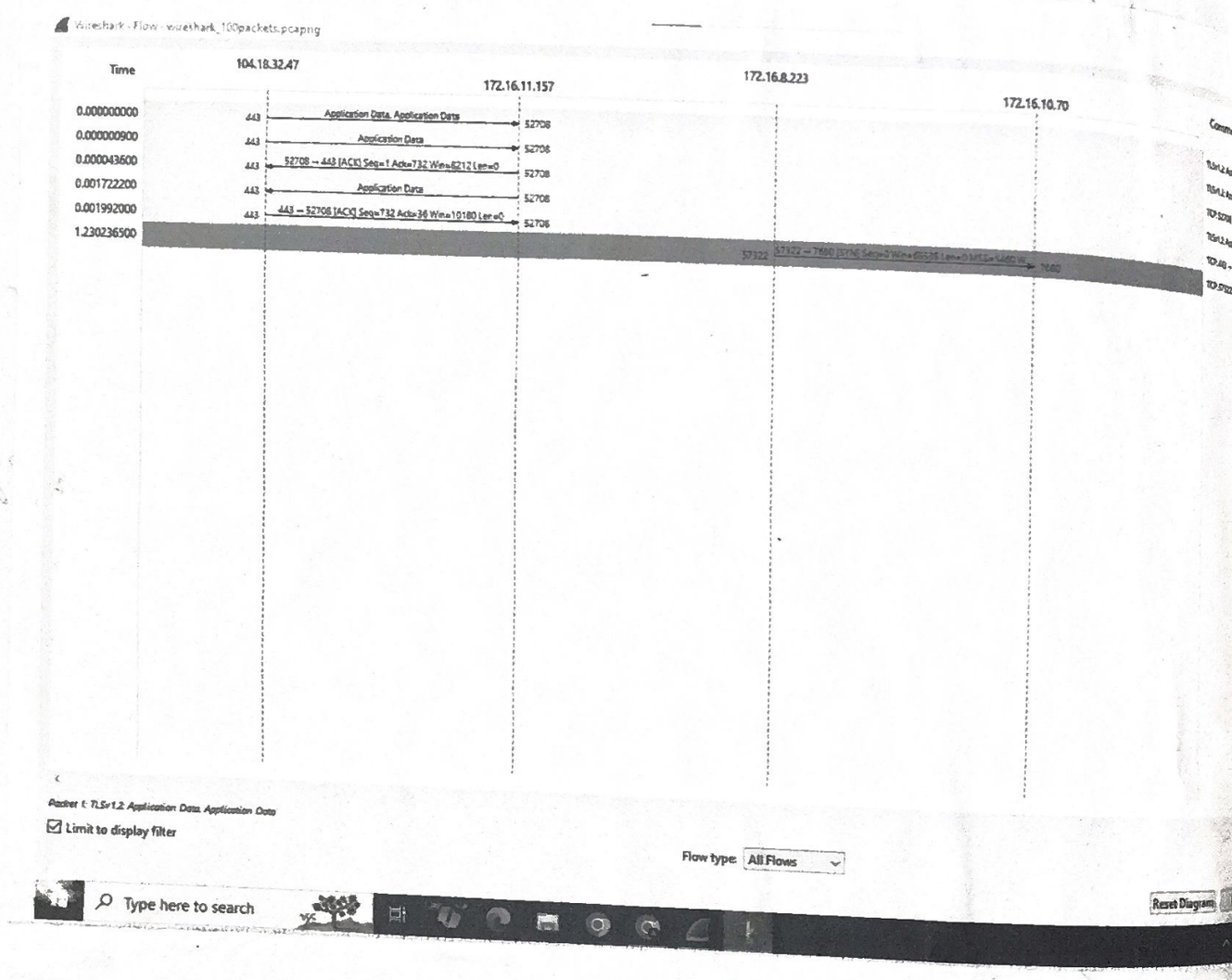
18) switch to simulation mode

19) Send a ping between two devices and play controls to step through the packet as it travels hop-by-hop, MAC-forwarding.

7) observation:

All four PC's are successfully connected in a star Topology and assigned unique static IP addresses with same subnet. Ping tests between all devices, confirms end-end connectivity. Confirming LAN was functioning correctly for resource sharing.





Result:

Thus, setting up a LAN and designing it is implemented successfully.